K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

SCHOOL OF ENGINEERING & TECHNOLOGY

LAB FILE

ENTERPRISE DIGITAL TRANSFORMATION USING CLOUD COMPUTING

Lab Code: ETMCCC273

Submitted To :

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MCA Sem 2

EXPERIMENT - 3: Launching and configuring an Amazon EC2 instance on AWS, installing Nginx, and hosting a webpage accessible through a public browser using User Data Script.

Objective

The objective of this experiment is to learn how to launch, configure, access, and manage an Amazon EC2 instance using the AWS Management Console. It also aims to understand how to install and configure a web server (Nginx) on an EC2 instance, create a simple HTML webpage, and access the hosted website through a public IP address. This experiment helps in understanding basic cloud computing concepts such as virtual machines, instance types, security groups, networking, and remote server management.

Theory

Amazon Elastic Compute Cloud (EC2) is a web service provided by AWS that allows users to create virtual servers in the cloud. These virtual machines can be configured with different operating systems, storage, and security settings according to user requirements.

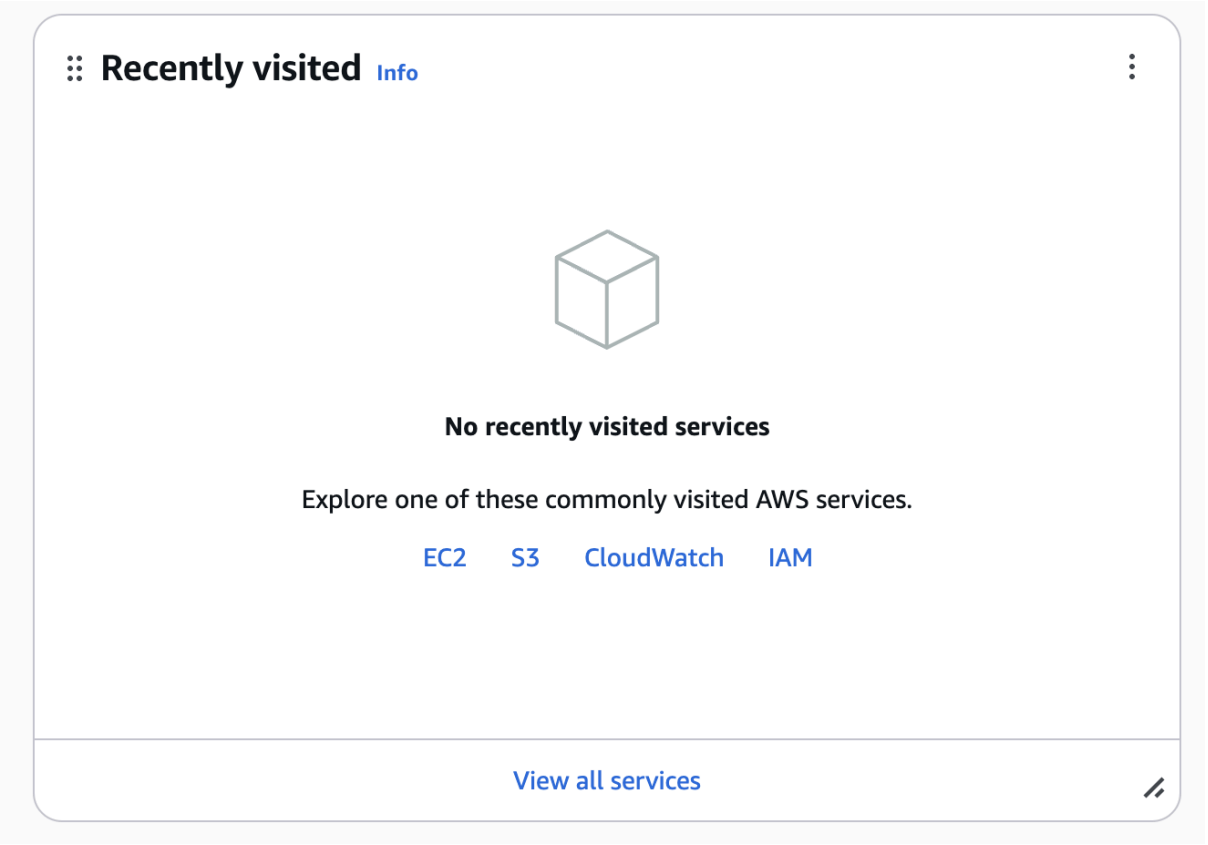
In this experiment, an EC2 instance was launched using the **Amazon Linux 2023 AMI**, which is a secure and optimized Linux distribution designed for AWS environments. The **t3.micro** instance type was selected because it provides 2 vCPUs and 1 GiB RAM and is suitable for small workloads and learning purposes under the AWS Free Tier.

After launching the instance, a **Security Group** was configured to allow SSH (port 22) for remote login and HTTP (port 80) for web traffic. The instance was accessed remotely using SSH, and the Nginx web server was installed to host a basic HTML webpage. The webpage files were stored in the default web directory `/usr/share/nginx/html`.

This experiment demonstrates the lifecycle of an EC2 instance, including launching, configuring networking rules, installing software, testing the server using commands like `curl localhost`, accessing the website via a public IP address, and understanding how cloud resources are managed efficiently.

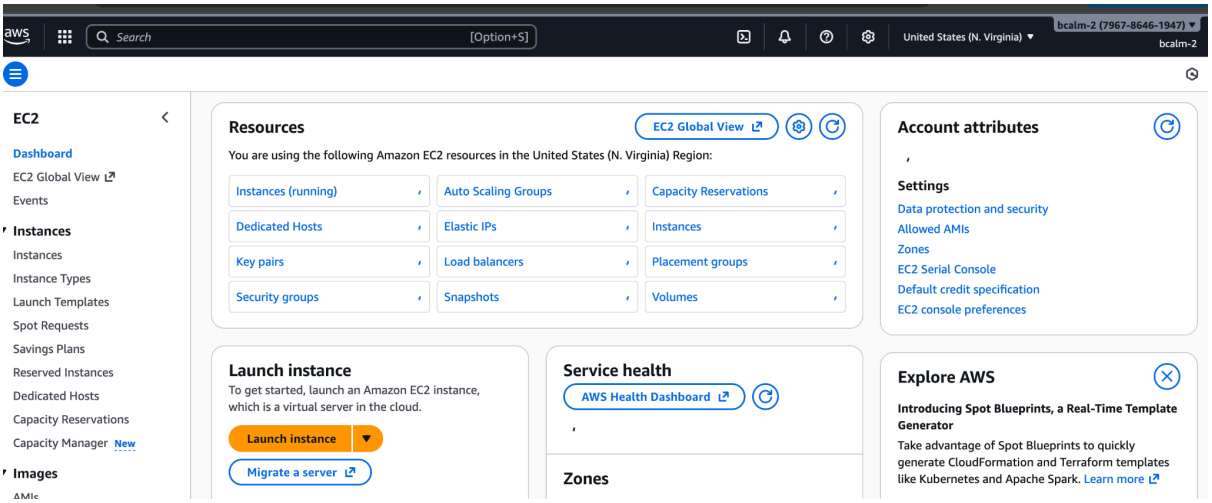
Steps Performed:

Step 1: Logged in to my AWS account using the AWS Management Console.

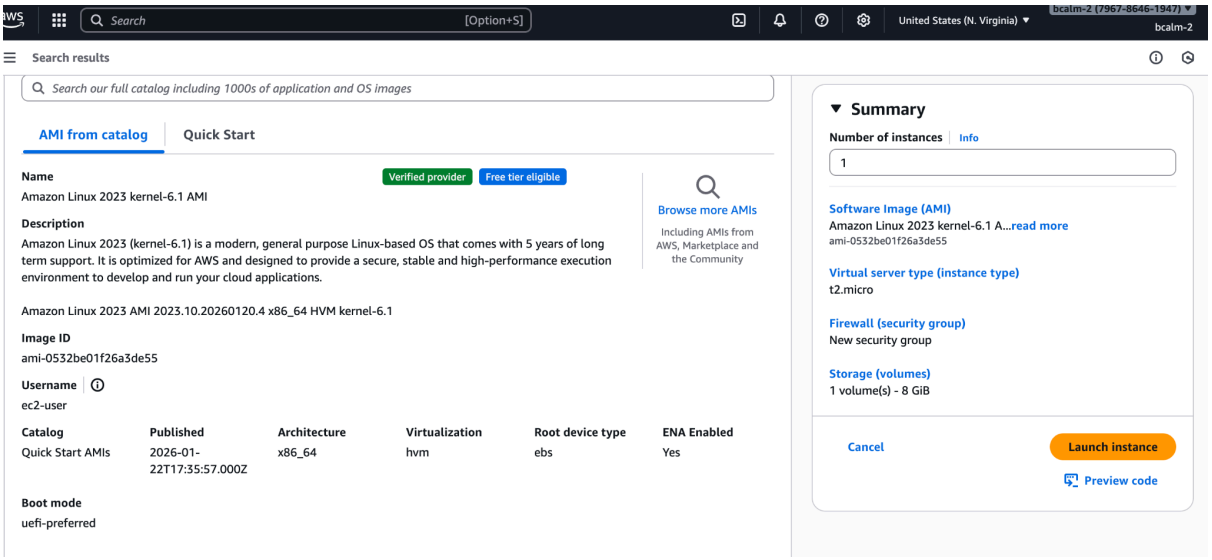


Step 2:

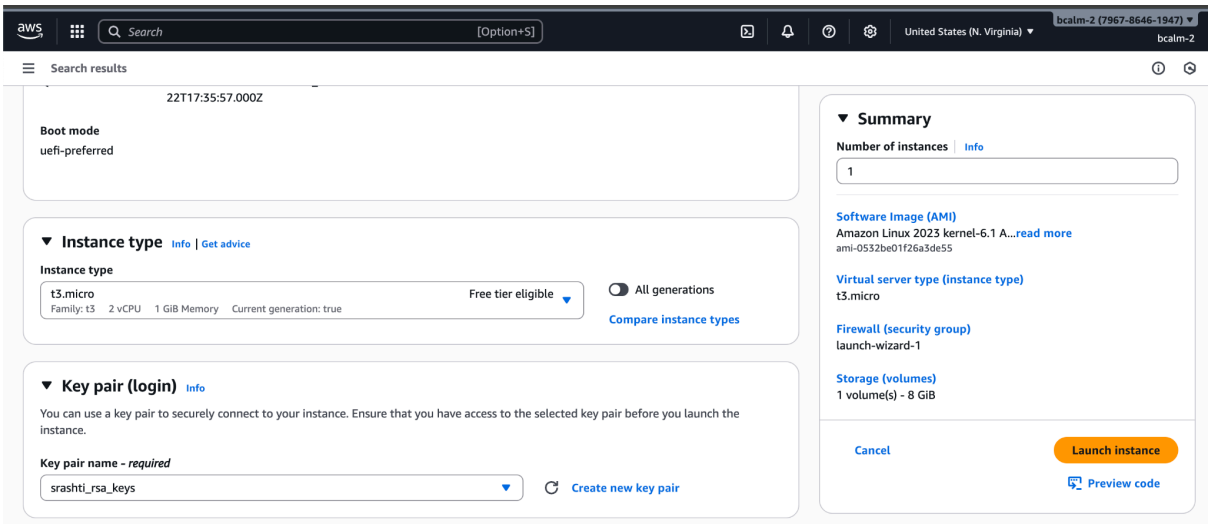
- Open the EC2 service from the AWS services list.
- Clicked on the Launch Instance button to create a new EC2 instance.



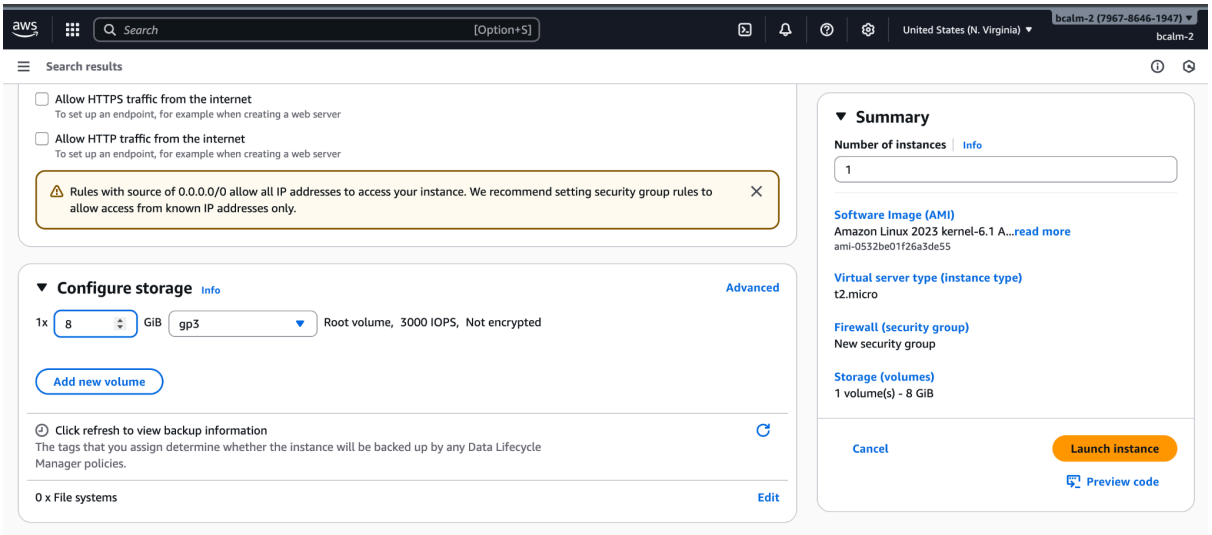
Step 3: Selected Amazon Linux 2023 (kernel 6.1) as the Amazon Machine Image (AMI).



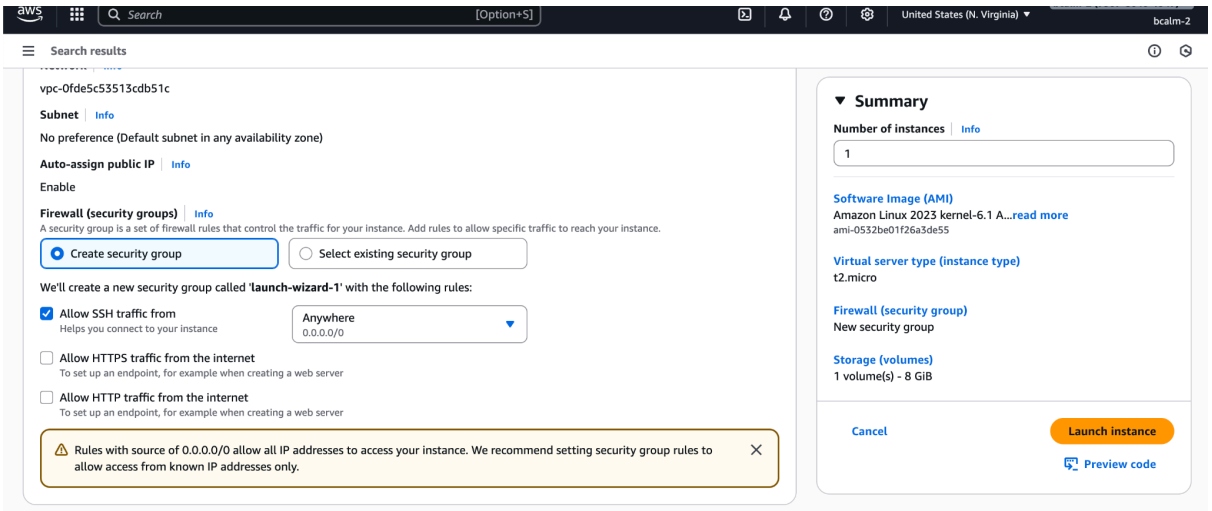
Step 4: Choose the instance type t3.micro, which provides basic CPU and memory for small workloads.



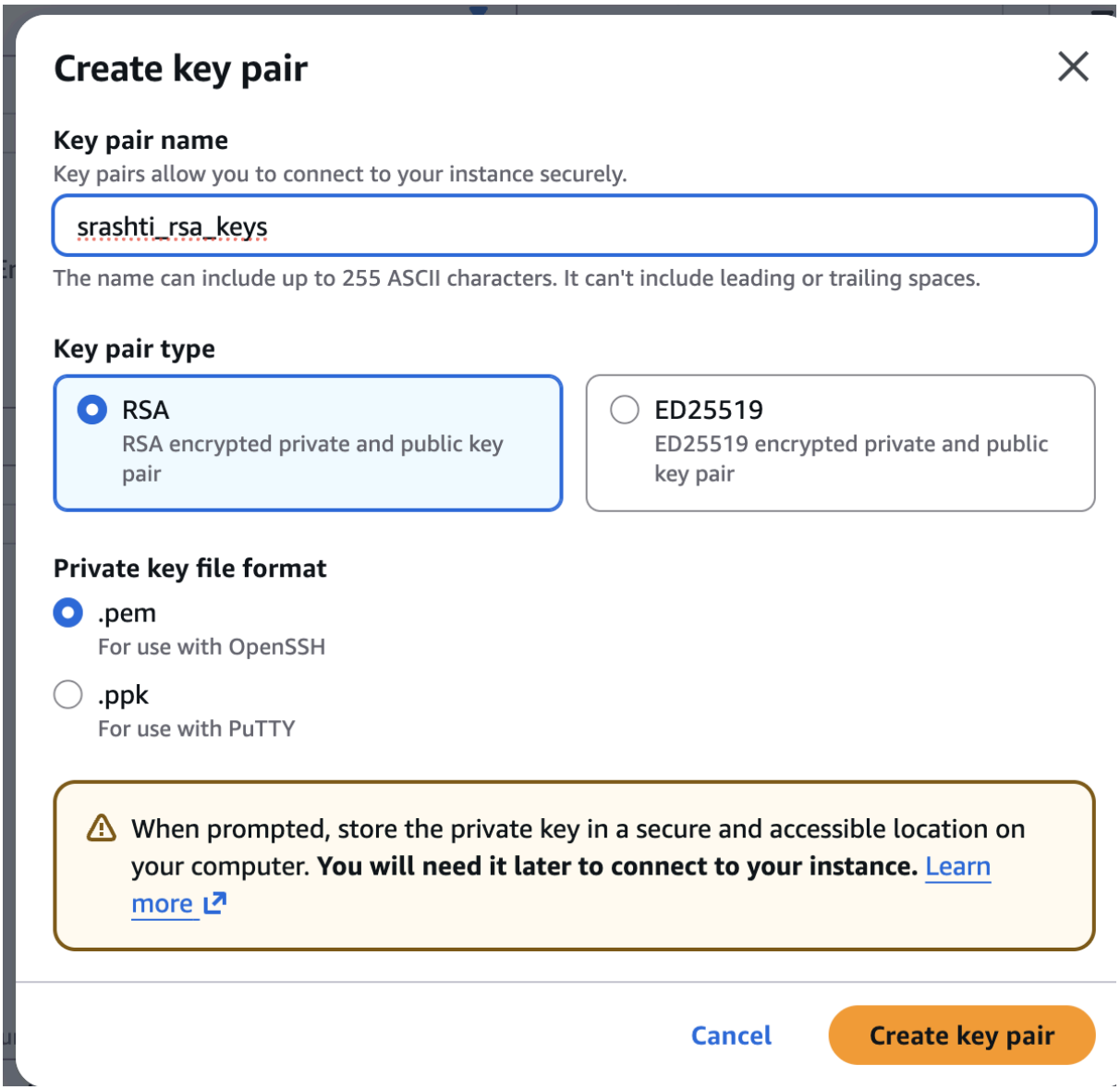
Step 5: Configure the storage size as 8 GiB.



Step 6: Set up the security group and allow **SSH access from anywhere** to connect to the instance..



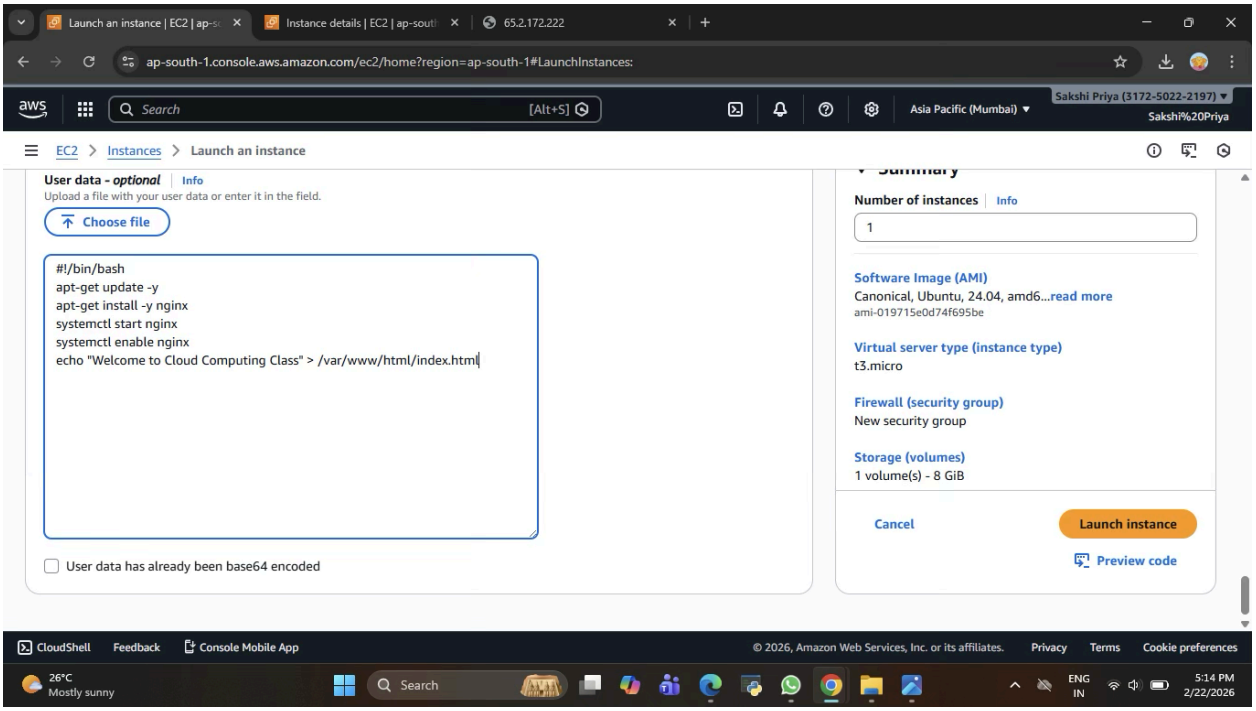
Step7: Create a new RSA key pair and download it for secure login.



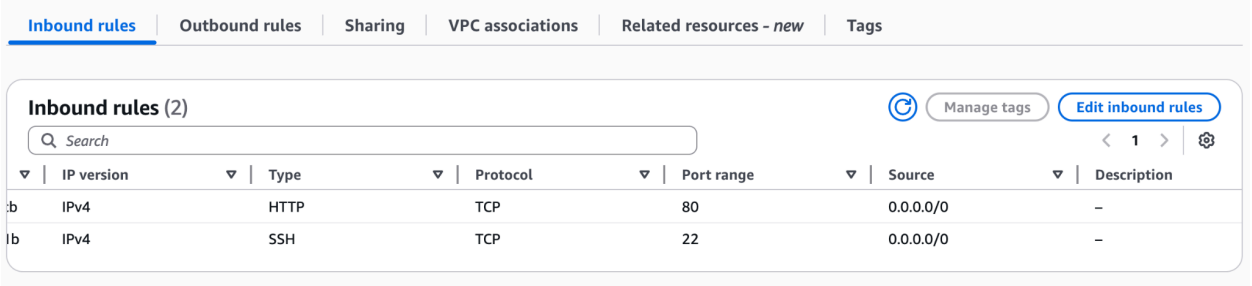
Step 8: Added Data user script and gave instructions to install nginx and update the landing index.html.

In the **User Data** section, added the following script:

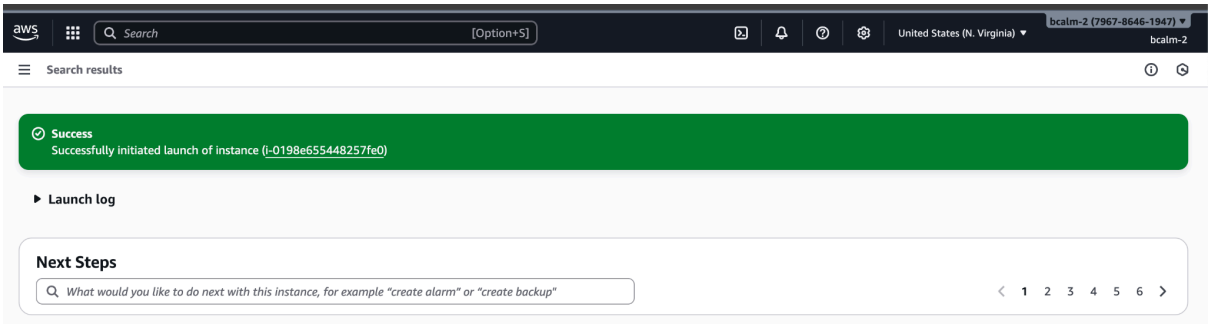
```
#!/bin/bash
apt-get update -y
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
echo "Welcome to Cloud Computing Class" > /var/www/html/index.html
```



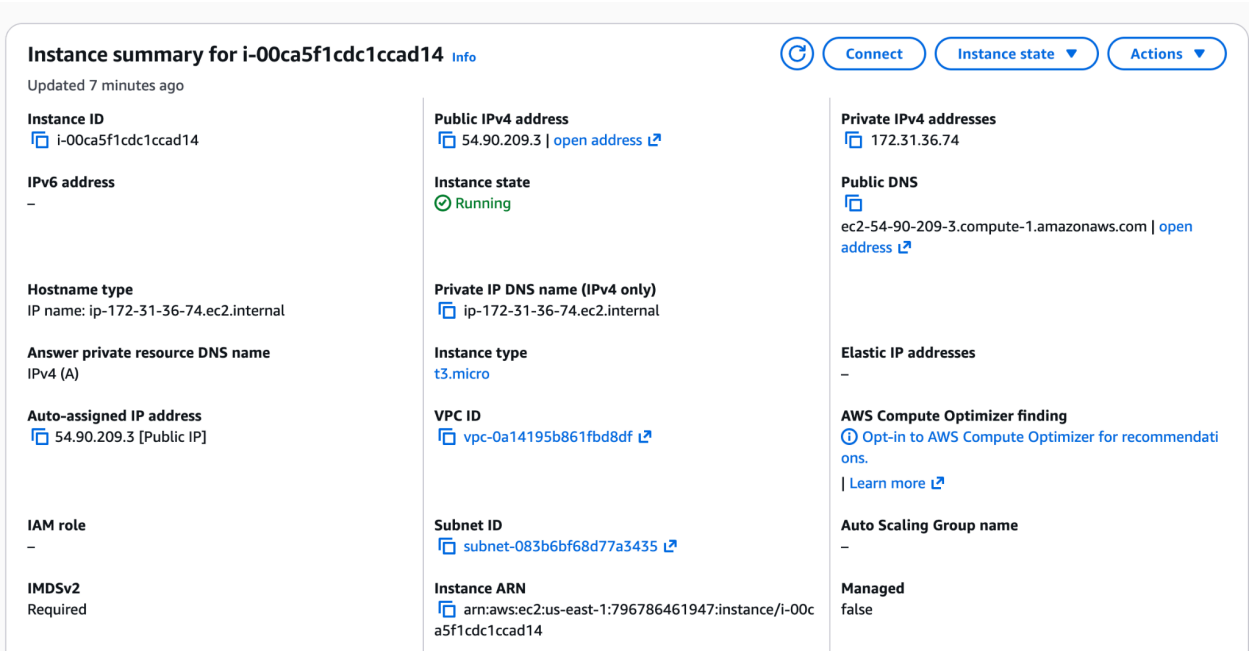
STEP 9: Configured the Security Group inbound rules to allow HTTP traffic so that the EC2 instance could accept incoming requests from a public browser.



Step 10: Click on **Launch Instance** and wait until the EC2 instance was created successfully.



Step 11: Verify the instance details and configuration after launch.



STEP 12: Verified the deployment by accessing the EC2 public IP address through a web browser to confirm that the hosted webpage was accessible.

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Result:

The Amazon EC2 instance was successfully launched and configured using the AWS Management Console. The Nginx web server was installed and started on the Amazon Linux 2023 instance. A custom HTML webpage was created in the default Nginx directory and verified using the curl localhost command. After configuring the Security Group inbound rules to allow HTTP traffic (port 80), the hosted webpage was successfully accessed through a public browser using the EC2 public IP address.

Precautions:

1. Ensure that the correct **Security Group inbound rules** are configured, especially allowing HTTP (Port 80) and SSH (Port 22).
2. Always use `http://` instead of `https://` unless SSL is configured on the server.

3. Verify that the EC2 instance has a **public IPv4 address** before accessing it from the browser.
4. Use proper permissions (sudo access) while modifying files inside the Nginx web directory.
5. Restart the Nginx service after making configuration changes to apply updates.
6. Keep the private key (.pem file) secure and never share it publicly.